Credit Risk Analytics

Data taken from :
https://www.kaggle.com/datasets/laotse/credit-risk-dataset?resource=download

Data Description:

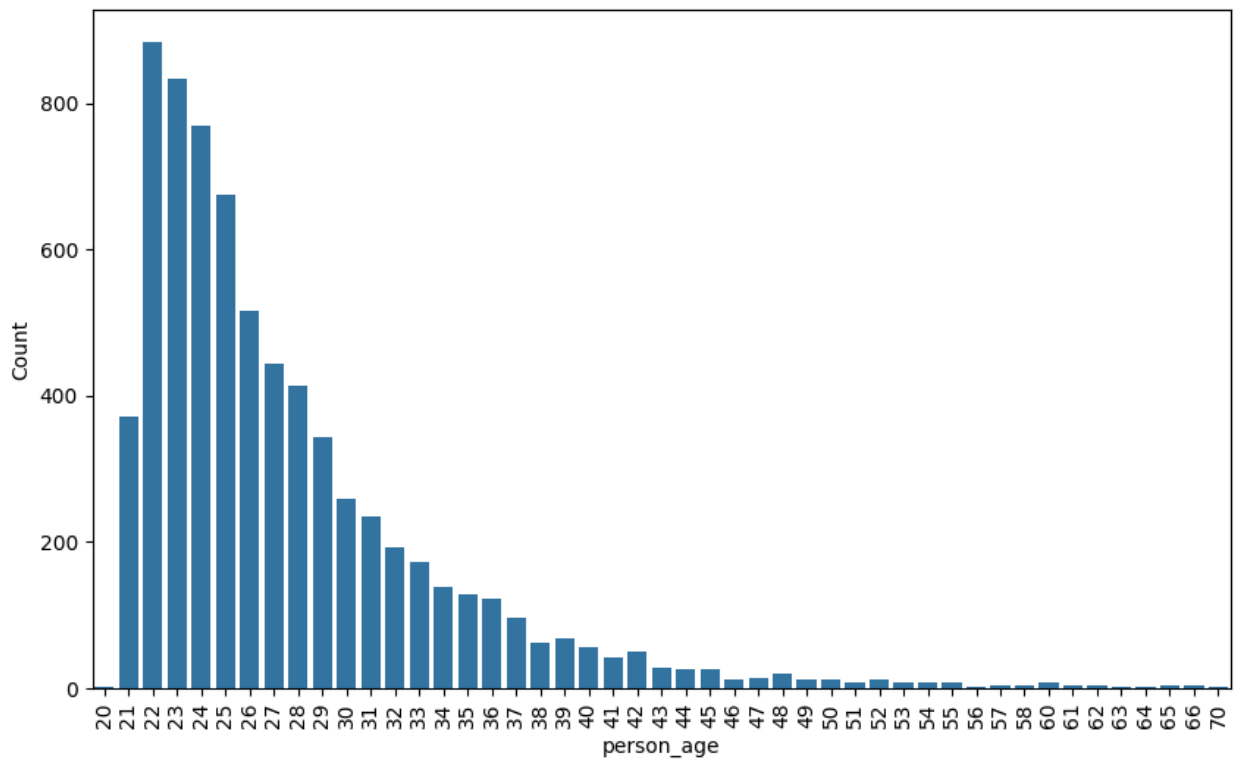
Feature Name -- Description

1. person_age -- Age
2. person_income -- Annual Income
3. person_home_ownership -- Home ownership
4. person_emp_length -- Employment length (in years)
5. loan_intent -- Loan intent
6. loan_grade -- Loan grade
7. loan_amnt -- Loan amount
8. loan_int_rate -- Interest rate
9. loan_status -- Loan status (0 is non default 1 is default)
10. loan_percent_income -- Percent income
11. cb_person_default_on_file -- Historical default
12. cb_preson_cred_hist_length -- Credit history length

Who is most likely to default?
Business Question: **Which customer segments have the highest probability of loan default?**

Does age affect default risk?
Question: **Are younger borrowers more likely to default compared to older customers?**

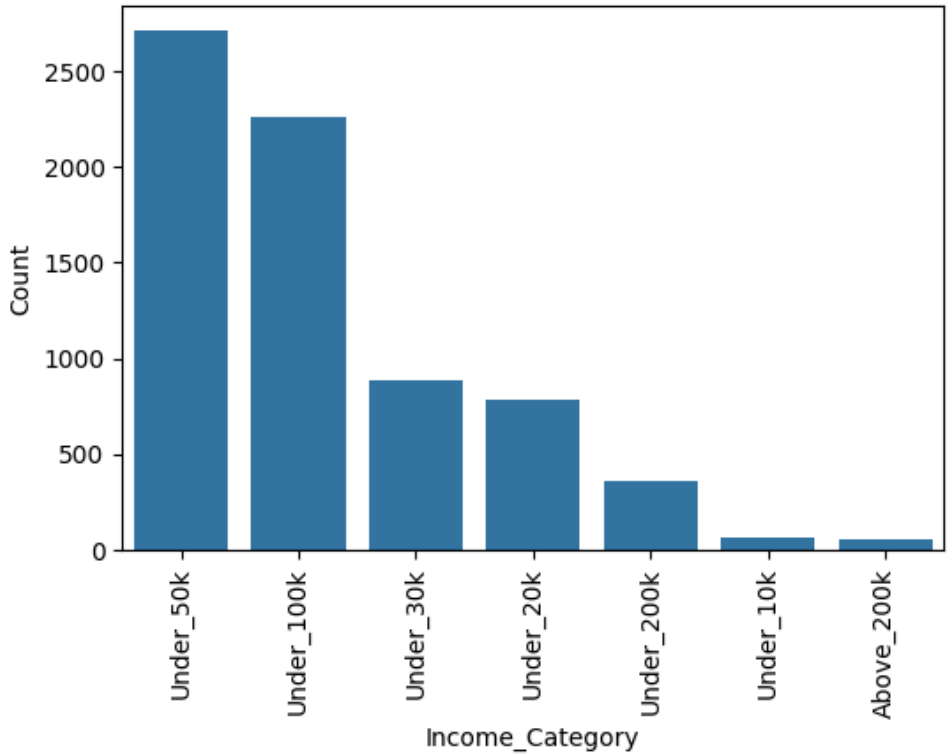
	person_age	Count
0	22	884
1	23	833
2	24	769
3	25	675
4	26	517



Age seem to have a strong correlation with Default Rates with younger audience defaulting more.
And Underwriting rules can be designed by taking this into account (age-based underwriting rules).

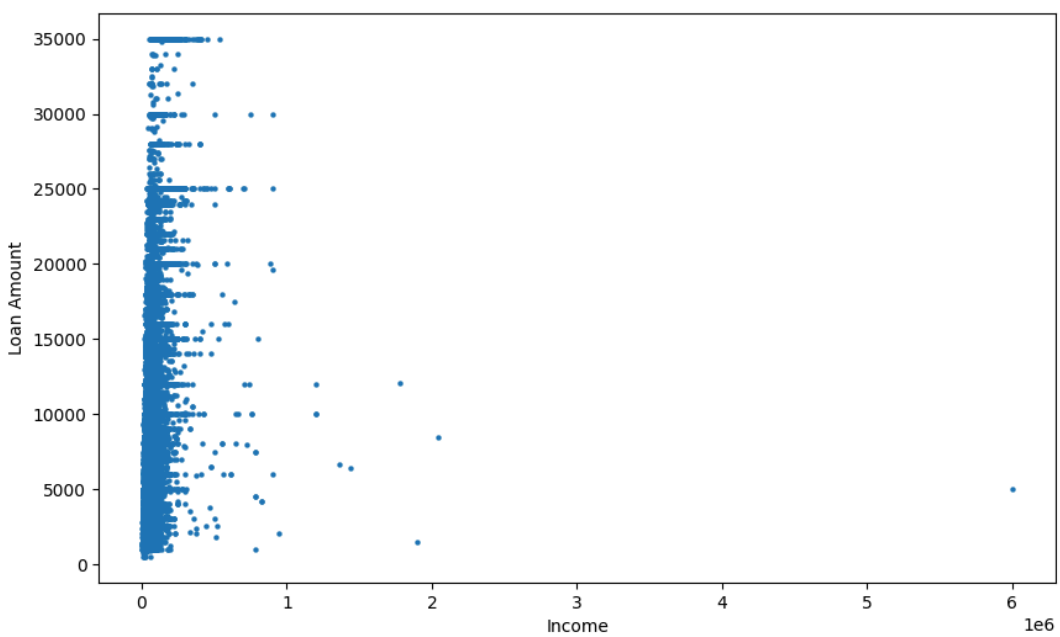
How does income impact loan default?
Question: **Is there a threshold income below which default rates sharply increase?**

	Income_Category	Count
0	Under_50k	2712
1	Under_100k	2261
2	Under_30k	881
3	Under_20k	785
4	Under_200k	354
5	Under_10k	61
6	Above_200k	54



1. Between 30k to 50k lies most of the Defaulters which is neither too low income nor too high. And then comes the Category >50k and <100k.
2. Therefore most Defaulters lie in bracket – '>30k and <100k'. That Category could have strong Underwriting Experience.
3. Also there are some cases of >200k who are defaulting. It's important to understand the amount for which they are defaulting as they are a high income category so loan amount could also be high which makes it much more risky even though frequency is few.
4. Also It is important to understand the count of total people in each of these categories so to understand the percentage of them defaulting and if there are ones(categories) which are defaulting most of the time.

Notably the max amount of loan is 35000 for income category >200k which has been taken by 9 persons.
Let's check overall income vs loan amount to get an idea whether loan amount is severely correlated with income or not.



It doesn't really show a strong correlation between Income and Loan Amount. Therefore High Income people need not have a strong Underwriting Experience.

Understanding Total Candidates in each category who have taken loan:

	Income_Category	Count_Defaulters	Total_Count	% Defaulted
0	Under_50k	2712	9635	28.147379
1	Under_100k	2261	14707	15.373632
2	Under_30k	881	2625	33.561905
3	Under_20k	785	970	80.927835
4	Under_200k	354	4031	8.781940
5	Under_10k	61	74	82.432432
6	Above_200k	54	539	10.018553

This completely changes the story.

"It now makes sense to have much stronger Underwriting for under 10k and under 20k categories as they are almost 80% of times defaulting."

Also we have kind of seen the Loan Amount is kind of not correlated with Income, so even small income candidates might have been taking larger loans. Therefore the above statement makes sense.

Also as income grows, % of defaulters are almost decreasing strongly.

Does home ownership reduce credit risk?
Question: **Do homeowners default less than renters?**

	person_home_ownership	perc_Defaulter	Total_Count	Count_Defaulter
0	RENT	31.569987	16446	5192
1	OTHER	30.841121	107	33
2	MORTGAGE	12.570663	13444	1690
3	OWN	7.469040	2584	193

1. Above Analysis can be useful in Risk Scoring Models.
2. Percent of people defaulting is highest in the RENT category including the count of them as well is the highest. Therefore a stronger Underwriting must need to be present for the ones with Home Ownership as RENT.
3. Secondly Countwise the second highest ones are the ones with MORTGAGE, however the percent of them is much lower than the RENT category, so there can be leniency present to them compared to RENT category.

Are people with longer employment histories less likely to default?
Business value: Helps set minimum employment requirements.

	Employment_Category	perc_Defaulter	Total_Count	Count_Defaulter
0	Above or equal to 50	50.000000	2	1
1	NaN	31.508380	895	282
2	Under or equal to 2	27.049407	10869	2940
3	Under_40	25.000000	8	2
4	Under_30	24.647887	142	35
5	Under_5	20.631912	6330	1306
6	Under_10	18.274719	10862	1985
7	Under_15	16.305916	2772	452
8	Under_20	15.000000	700	105
9	Under_50	0.000000	1	0

1. No clear pattern , however the ones with under 2 years of employment and unknown history constitutes the highest percentage in terms of defaulting. Also the number of defaulters is highest for the Under_2 category.
2. So at least 2 years of employment could drastically reduce the chance of defaulting as it will mitigate both NAN category and Under 2 year category.

Loan Characteristics Analysis:

Which loan intents have the highest default rate?
Question:

1. **Do loans taken for education, personal use, or medical reasons default more often?**

Some loan purposes may need stricter approval criteria.

	loan_intent	perc_Defaulter	Total_Count	Count_Defaulter
0	DEBTCONSOLIDATION	28.587874	5212	1490
1	MEDICAL	26.700708	6071	1621
2	HOMEIMPROVEMENT	26.102635	3605	941
3	PERSONAL	19.887702	5521	1098
4	EDUCATION	17.216798	6453	1111
5	VENTURE	14.810282	5719	847

1. Personal, Education, Venture have low percent Defaulters.
2. Homeimprovement have low loan taken count but high percent of defaulters. So this could be the area where the Sales pitch step could take a backstep and focus of Sales could be focussed on areas of Venture, Education and Personal in order.
3. DebtConsolidation, and Medical both have high percent defaulters but their loan taken count is also high. Especially Medical have much higher loan_taken count but percent default is slightly lower than DebtConsolidation. So these could be the areas where focus should be on stricter loan approval mechanism and Sales pitch could be as it is or improved.
4. Also Medical could be the area where focus can be shifted overall as it could help garner much revenue and with strict approval mechanism if default rate is decreased then Growth could be exponential.

Validating whether credit grading system is effective

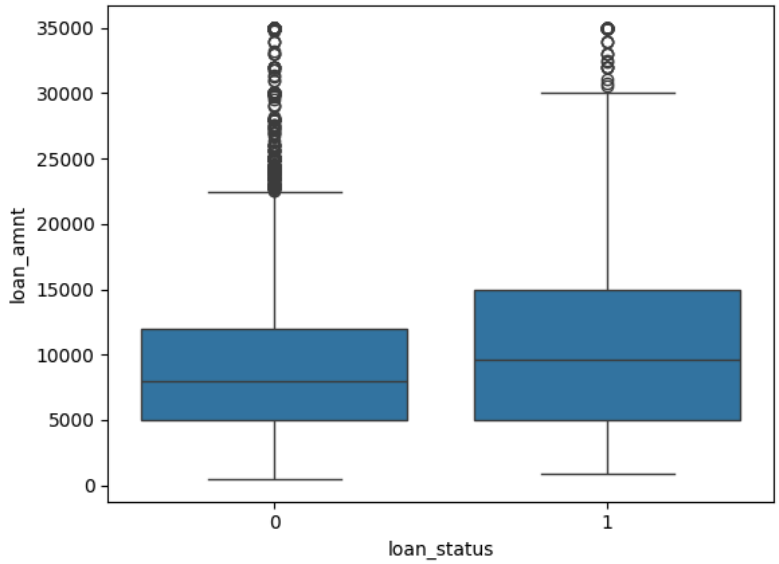
	loan_grade	perc_Defaulter	Total_Loan_Amount	Total_Count	Count_Defaulter
0	G	98.437500	1100525	64	63
1	F	70.539419	3546875	241	170
2	E	64.419087	12450875	964	621
3	D	59.045780	39339350	3626	2141
4	C	20.733973	59503125	6458	1339
5	B	16.275954	104462800	10451	1701
6	A	9.956389	92027750	10777	1073

Loan Grade B has more default rate than A which is expected but is also contributing to more Loan Amount given which is where the focus should be and so the focus should be on producing more loan (especially loan amount) which is grade A.

In terms of Loan Count A has more loans but Total amount of B is exceeding A which means there are some high end loans given in Category B, and so the focus should be is on converting those loans to grade A.

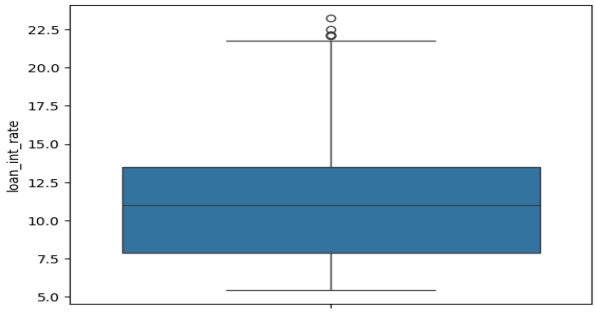
Or simply put, customers should be found with grade A attributes which are requiring high end loans since A grade contributes to small default rates.

Are larger loans riskier?
Question: **Does the probability of default increase as loan amount increases?**



Clearly the Median and the 75th percentile and whiskers, all three are higher in case of Defaulted loans which tells, people with high end loans seems to tend to default more.

Also notably there are people in both defaulters and non defaulters category who have taken extreme loans reaching up to 35k $ or more.

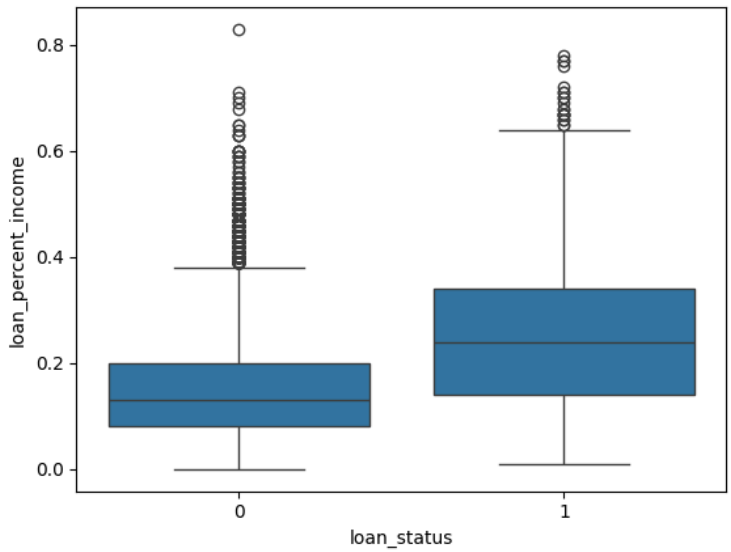


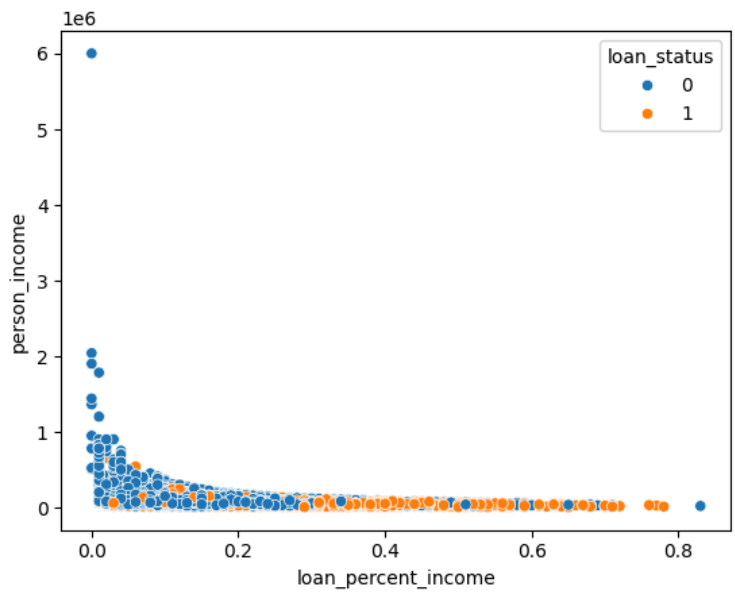
	Loan_int_Category	perc_Defaulter	Total_Count	Count_Defaulter
0	Above or equal to 20	85.135135	74	63
1	Under_20	66.059818	769	508
2	Under_17	59.593023	1032	615
3	Under_16	51.724138	1566	810
4	Under_15	42.474130	2126	903
5	Under_14	22.260991	2866	638
6	NaN	20.667522	3116	644
7	Under_13	20.081061	2714	545
8	Under_12	16.862861	3398	573
9	Under_11	16.550117	3432	568
10	Under_10	14.507254	1999	290
11	Under_9	12.856312	1719	221
12	Under_8	10.816175	4031	436
13	Under_5	7.863065	3739	294

1. Clearly Loan Default is increasing as interest rate is increasing.
2. The number of people who have taken the loan is invariably distributed across different loan interest rate categories with most loans lying between interest rate 10 and 15.

Debt Burden Analysis

What is the relationship between income utilization and default?
Question: Do borrowers spending more than a certain percentage of their income on loan payments default more?





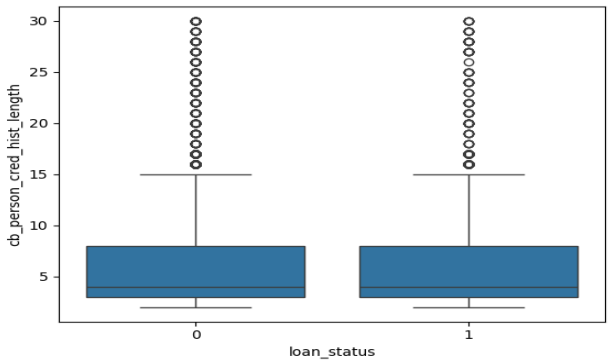
The above graph tells when loan_percent_income increased, Defaulters also increased.
Let's check that in more detail:

	Loan_percent_income_Category	perc_Defaulter	Total_Count	Count_Defaulter
0	Under_0.8	81.818182	11	9
1	Under_0.6	77.454545	275	213
2	Under_0.7	73.469388	49	36
3	Under_0.5	72.572573	999	725
4	Under_0.4	61.762712	2950	1822
5	Under_0.3	20.888954	6817	1424
6	Under_0.2	14.734994	12528	1846
7	Under_0.1	11.540610	8951	1033
8	Under_0.9	0.000000	1	0

1. As Loan_percent_income increased, percent Default also increased, which tells the people with more constraints on paying back loans tend to default more.
2. But notably the total number of people in those brackets are also invariably low which tells about the Bank not providing loans to those people in larger numbers or might have taken this constraint in account already.
3. The safest bracket is Under 0.1 category. Also though the Under_0.9 category has 0 percent default rate but that only has 1 candidate, so data is insufficient for that bracket.

Does credit history length reduce default risk?

Question: Are customers with longer credit histories more reliable?

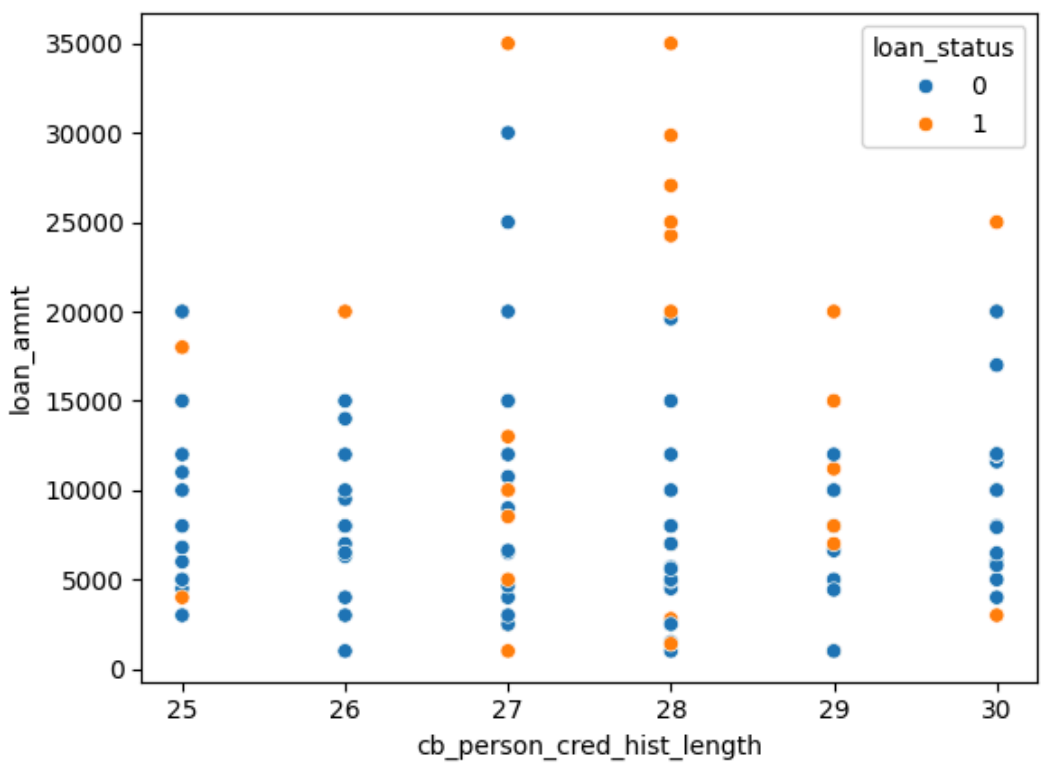


	Credit_history_length	perc_Defaulter	Total_Count	Count_Defaulter
0	Between_25_and_31	28.571429	119	34
1	Between_20_and_25	24.000000	125	30
2	Between_2_and_3	23.570830	5965	1406
3	Between_4_and_5	22.312236	5925	1322
4	Between_3_and_4	22.278311	5943	1324
5	Between_15_and_20	20.898876	1335	279
6	Between_6_and_7	20.786214	1857	386
7	Between_7_and_10	20.709021	5698	1180
8	Between_5_and_6	20.627326	1881	388
9	Between_10_and_15	20.332173	3733	759

1. There is no clear pattern in defaulters varied from non defaulters based on Credit history length.
2. Also the people with the most credit history length have the highest default rate which is very different from natural behavioral characteristics of users "where more credit history should make them reliable".

So let's study these users a bit:

	cb_person_cred_hist_length	perc_Defaulter	Total_Loan _Amount	Total_ Count	Count_Defaulter
0	29	42.857143	119250	14	6
1	27	36.363636	249050	22	8
2	28	33.333333	295175	27	9
3	25	27.777778	186300	18	5
4	30	22.727273	231625	22	5
5	26	6.250000	140300	16	1

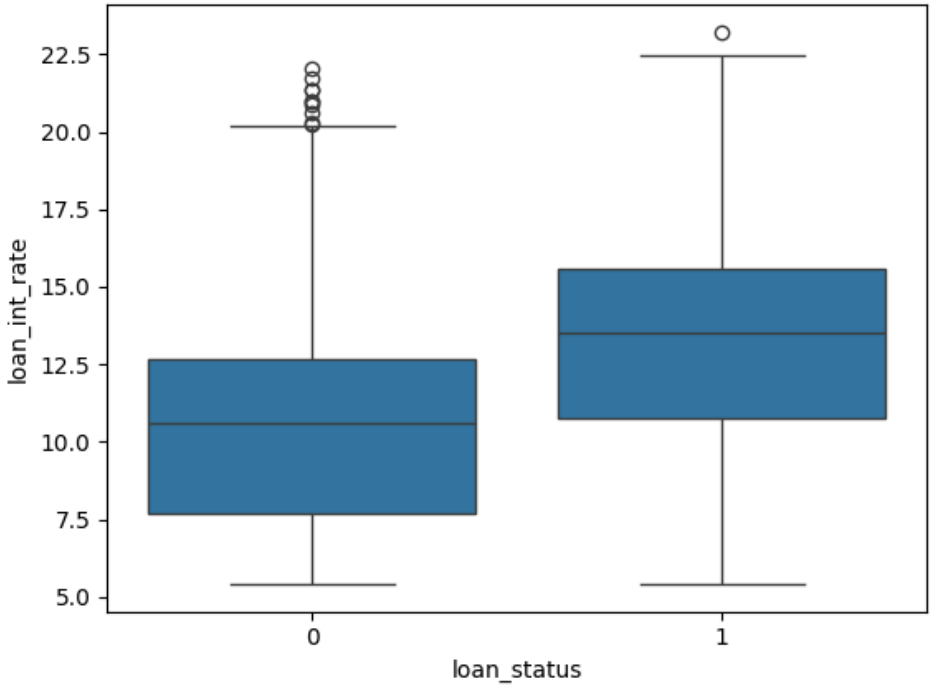


1. No clear pattern among these users, though Above 28 - credit history length combined with high income - 20k or more seems to default usually.

Policy Optimization
Should the bank reject customers with prior defaults?
Question: Is the default rate too high for customers who have defaulted previously?

1. 37.8 % of all defaulters have defaulted again compared to 18.3 % of Non Defaulters defaulting. This seems like a big jump and provides an area to strictly look at to frame policy whether to provide loans to such people or not altogether. I suggest only in rare circumstances these people should be given loans.
2. Also their defaulting history needs to be carefully analyzed (circumstances they have been in) before putting them in a bucket of Defaulters.

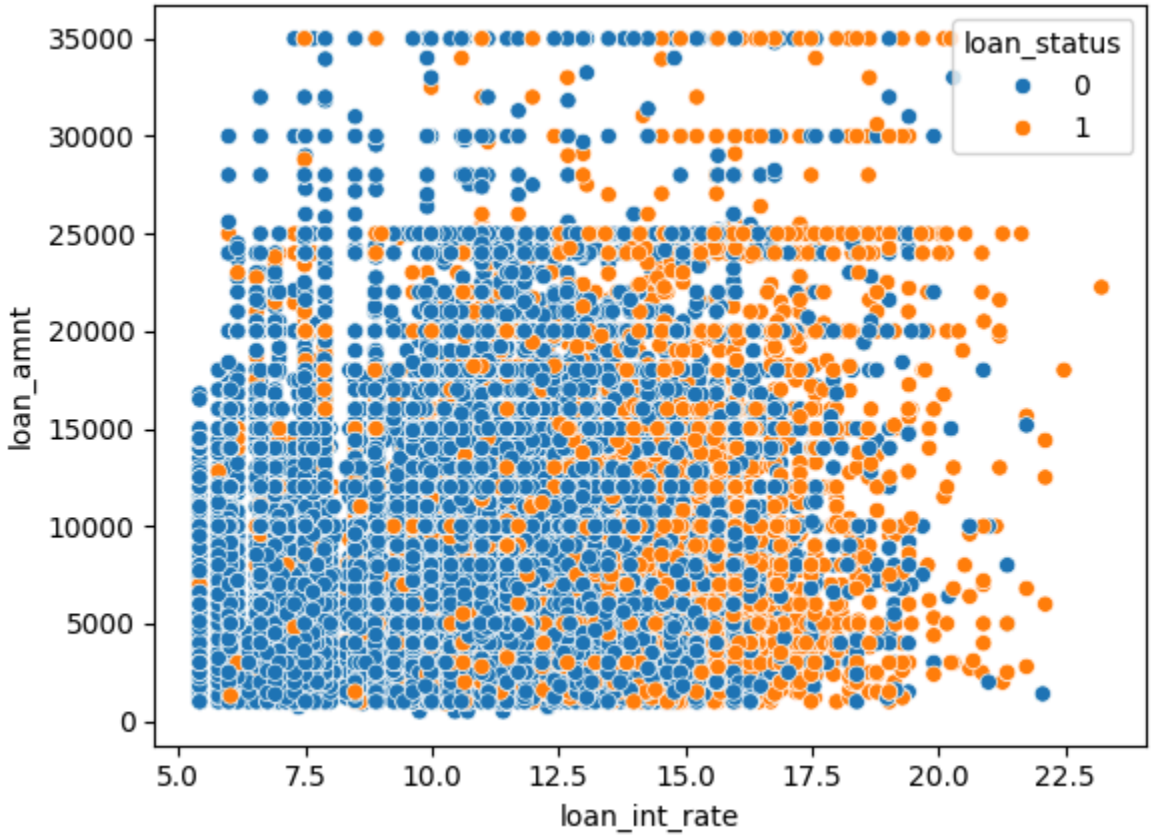
Pricing and Profitability
Is the bank pricing risk correctly?
Question: Are customers with higher default risk actually being charged higher interest rates?



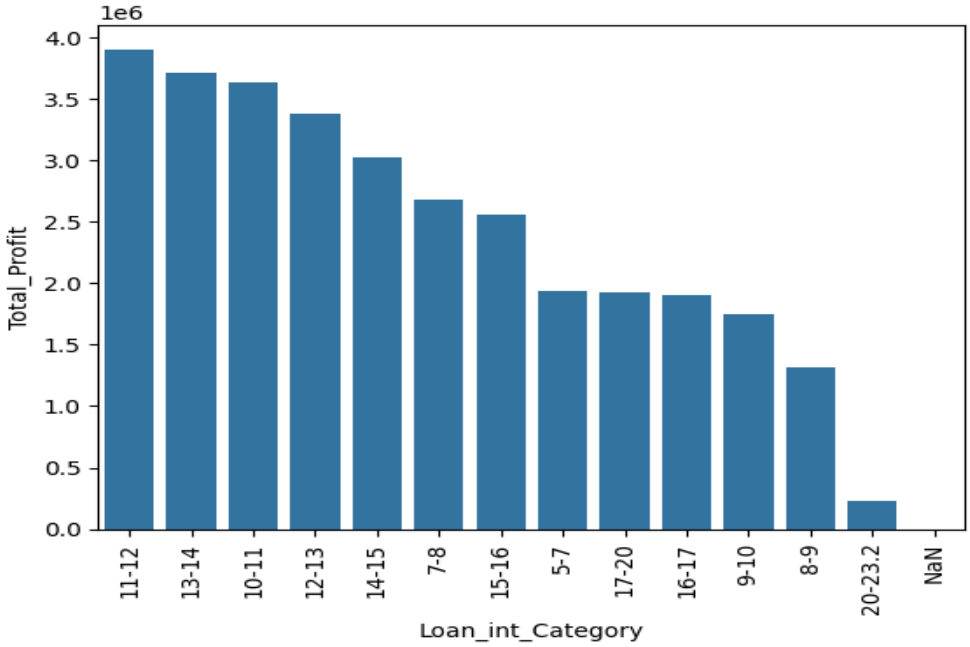
Customers with higher default risk actually being charged higher interest rates in general.

What interest rate range balances risk and profitability?

Question: At what interest rate does the bank maximize returns without increasing default too much?



	Loan_int_Category	Total_loan_amnt	perc_Defaulter	Total_Profit
0	11-12	33801500	16.862861	3.906420e+06
1	13-14	27470900	22.260991	3.713096e+06
2	10-11	34247625	16.550117	3.630741e+06
3	12-13	26865525	20.081061	3.383930e+06
4	14-15	20844875	42.474130	3.029867e+06
5	7-8	35278675	10.816175	2.676916e+06
6	15-16	16505825	51.724138	2.562623e+06
7	5-7	30916000	7.863065	1.932621e+06
8	17-20	10629075	66.059818	1.923758e+06
9	16-17	11569750	59.593023	1.902219e+06
10	9-10	18011500	14.507254	1.751406e+06
11	8-9	15196175	12.856312	1.311936e+06
12	20-23.2	1077075	85.135135	2.244828e+05
13	NaN	30016800	20.667522	NaN



1. Maximum Profit lies 10-15 interest rate and 11-12 being maximum within that.
2. So this could be the interest rate that the Bank can target for maximum profit and lowest risk based on trends.
   And assuming the bank has certain criteria for someone to come under this interest rate category. Those are the segment of customers banks should target and put most of their Marketing and Sales efforts on.

FINAL ANALYSIS:

Portfolio Risk Management

1. **Which combination of factors produces the highest default risk?**
   Question: What combination of income, age, and loan amount creates the highest risk?
   Business value: Identify high-risk customer profiles.

   The highest default risk occurs for borrowers with:
   - Low income
     - Especially Under 10k and Under 20k income groups
     - Default rates ~80%+
   - Younger age group
     - Younger borrowers show higher default tendency
   - Higher loan amounts
     - Defaulted customers have higher median and upper loan amounts
   - High loan burden (loan_percent_income)
     - Higher % of income spent → significantly higher defaults

   Final High-Risk Profile

   Young + Low Income (especially <20k) + High Loan Amount + High Loan-to-Income Ratio

2. **What are the safest borrower profiles?**
   Question: Which customer segments consistently show the lowest default rates?

   Safest borrowers consistently show:
   - Higher income groups
     - Especially Under 200k (8.7%) and Above 200k (~10%) default rates
   - Home ownership
     - OWN (7.46%) and MORTGAGE (~12.5%) have lowest default rates
   - Low loan_percent_income
     - <10% income utilization → ~11.5% default rate (lowest stable group)
   - Longer employment history
     - >2 years employment reduces default significantly
   - Better loan grades (A, B)
     - Grade A ~10% default, lowest among all

   Final Safest Profile

   High Income + Homeowner (OWN/MORTGAGE) + Low Loan Burden (<10%) + Stable Employment (>2 years) + Good Credit Grade (A/B)

3. **What approval rules would minimize default?**
   Question: What combination of rules (income, credit history, employment) would reduce default the most?

   Rule 1: Income-Based Filtering

   - Apply strict underwriting for income <20k
   - Possibly reject or heavily restrict <10k group
   - Focus approvals on >50k income segments

   Rule 2: Employment Stability

   - Require minimum 2 years employment
   - Avoid:
     - Under 2 years
     - Missing employment history

   Rule 3: Loan Burden Control

   - Cap loan_percent_income ≤ 0.1–0.2
   - Avoid high burden segments (>0.4) due to extreme default rates

   Rule 4: Prior Default Screening

   - Customers with past defaults show:
     - ~37.8% repeat default rate vs ~18.3% otherwise
   - Recommendation:
     - Approve only in rare/exception cases

   Rule 5: Home Ownership Preference

   - Prefer:
     - OWN, MORTGAGE
   - Apply stricter rules for:
     - RENT category (highest defaults)

Rule 6: Loan Amount Control

- Limit high loan exposure for risky segments
- Since higher loan amounts → higher defaults

Final Optimal Approval Strategy

Approve borrowers with:
Income >50k + Employment >2 years + Loan ≤20% of income + No prior default + Home ownership (OWN/MORTGAGE)

Reject or restrict:
Low income + high loan burden + unstable employment + past defaults

Final Summary:

- **High risk = young, low income, high loan burden customers**
- **Safe segment = financially stable, low leverage, experienced borrowers**
- **Optimal policy = tight underwriting on income, employment, and past defaults + control loan-to-income ratio**